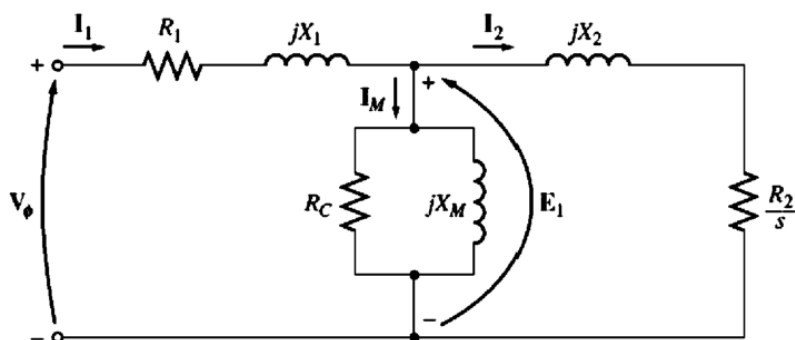


Exercises 4 – Induction Motor (IM)

1. A 220-V three-phase six-pole 50-Hz induction motor is running at a slip of 3.5 percent. Find:
 - a. The speed of the magnetic fields in revolutions per minute?
 - b. The speed of the rotor in revolutions per minute?
 - c. The slip speed of the rotor?
 - d. The rotor frequency in hertz?
2. A 50-kW 440-V 50-Hz two-pole induction motor has a slip of 6 percent when operating at full-load conditions. At full-load conditions, the friction and windage losses are 520 W, and stray losses are 500W. Find the following values for full-load conditions:
 - a. The shaft speed n_m ?
 - b. The load torque τ_{load} in newton-meters?
 - c. The induced torque τ_{ind} in newton-meters?
 - d. The rotor frequency in hertz?
3. A three-phase 60-Hz two-pole induction motor runs at a no-load speed of 3580 rpm and a full-load speed of 3440 rpm. Calculate the slip and the electrical frequency of the rotor at no-load and full-load conditions? What is the speed regulation of this motor? Given:

$$SR = \frac{n_{nl} - n_{fl}}{n_{fl}}$$
4. A 460-V, 25-hp, 60Hz, four-pole, Y-connected induction motor has the following impedances in ohms per phase referred to the stator circuit:



$$R_1 = 0.5\Omega \quad ; \quad R_2 = 0.2\Omega$$

$$X_1 = 1\Omega \quad ; \quad X_2 = 0.35\Omega \quad ; \quad X_M = 0.2\Omega \quad ; \quad R_C = 0\Omega$$

For a rotor slip of 2.2 percent at the rated voltage and rated frequency, find the motor:

- a. Mechanical shaft speed
- b. Stator current
- c. Power factor
- d. P_{conv}
- e. Efficiency